Fundamentals of Natural Language Processing

CS115B — Spring 2025

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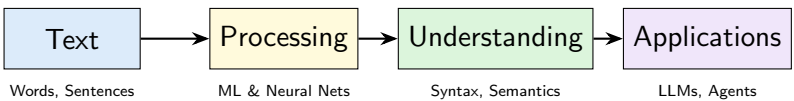
Department of Computer Science
Brandeis University

Spring 2025

Welcome to CS115B!

Fundamentals of Natural Language Processing

The second course in the Computational Linguistics MS sequence



Course Information

Logistics:

- **Course:** CS115B
- **Credits:** 4
- **Semester:** Spring 2025
- **Website:** `https://cs-114.org`

Textbooks:

- Eisenstein, *Natural Language Processing* (E)
- Jurafsky & Martin, *Speech and Language Processing* (JM)

Prerequisites:

- Linear algebra
- Differential and vector calculus
- Python programming
- NumPy, PyTorch familiarity

Tools:

- Python
- NumPy
- PyTorch
- Standard NN libraries

Course Description

This course covers **fundamental concepts and algorithms** in computational linguistics and natural language processing.

We will explore computational analysis **from the word level to the sentence level**:

- **Text classification**
- **Part-of-speech tagging**
- **Probabilistic statistical parsing**
- **Computational distributional lexical semantics**
- **Neural networks** for NLP

The **lab component** introduces necessary mathematical concepts and computational libraries.

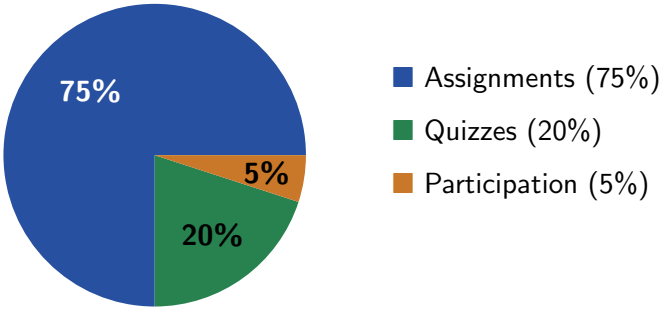
Note: If you do not know Python, you will need to learn quickly during the first week!

Learning Goals

Upon completion of this course, students will:

1. **Understand the fundamentals** of processing natural language using machine learning algorithms
2. Master key algorithms:
   - Naive Bayes and Logistic Regression classification
   - Perceptron, Multilayer Perceptrons, Structured Perceptrons
   - Recurrent Neural Network architectures
3. **Learn to create word embeddings** for language models and NLP applications
4. Become **proficient in NumPy** and familiar with standard neural network libraries
5. Understand modern **Transformer architectures and LLMs**

Grading



- **Assignments** are the core of the course—hands-on implementation
- **One midterm quiz** and **one final quiz**
- **Class participation** includes discussion and engagement

Workload: Expect **9+ hours/week** outside class (readings, assignments, exam prep)

Accommodations & Support

Disability Accommodations:
If you are a student with a documented disability on record at Brandeis University and wish to have reasonable accommodations made for you in this class, please see me **immediately**.

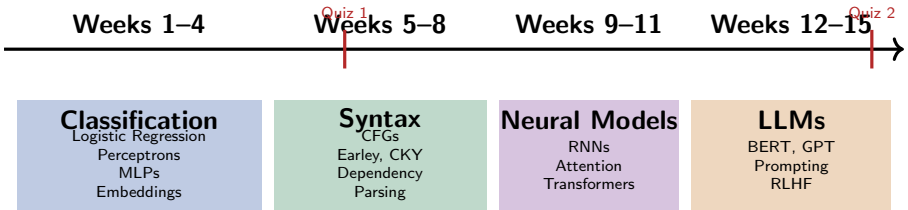
Office Hours:

- Check course website for current office hours
- Additional help available through course staff

Resources:

- Course website: `https://cs-114.org`
- Lab materials and assignments posted online
- Discussion forum for questions

The Arc of the Course



Big picture: We build from classical ML → neural networks → modern LLMs, always grounded in linguistic analysis.

Schedule: Foundations & Classification (Weeks 1–4)

Date	Topic	Readings
1/13	Two Cultures of Linguistics	E1
1/16	Two Cultures (Part 2); Logistic Regression	E2.1–2.2; 4.1; 4.3–4.4; JM4
1/20	Logistic Regression (Part 2)	E2.5–2.7; JM5
1/23	Logistic Regression (Part 3)	—
1/27	Evaluation; Perceptrons	E2.3–2.4; 2.8
1/30	Structured Prediction (Part 1)	—
2/3	Structured Prediction (Part 2)	—
2/6	Multilayer Perceptrons	E3.1–3.3; JM7; E7.1–7.5
2/10	Multilayer Perceptrons (Part 2)	JM6; E6.1–6.2; 6.4–6.5; JM3
2/13	Multilayer Perceptrons (Part 3)	—

Key skills: Classification, SGD, structured prediction, neural network basics.

Schedule: Semantics & Syntax (Weeks 5–8)

Date	Topic	Readings
2/24	Distributional Semantics (Part 1)	—
2/27	Distributional Semantics (Part 2)	—
3/3	Syntax	—
3/6	**Quiz 1**	—
3/10	Parsing (Earley)	—
3/13	Parsing (CKY)	—
3/17	Neural Parsing	—
3/20	Dependency Parsing	—
3/24	Dependency Parsing (continued)	—
3/27	Neural Dependency Parsing	E9.2; JM18

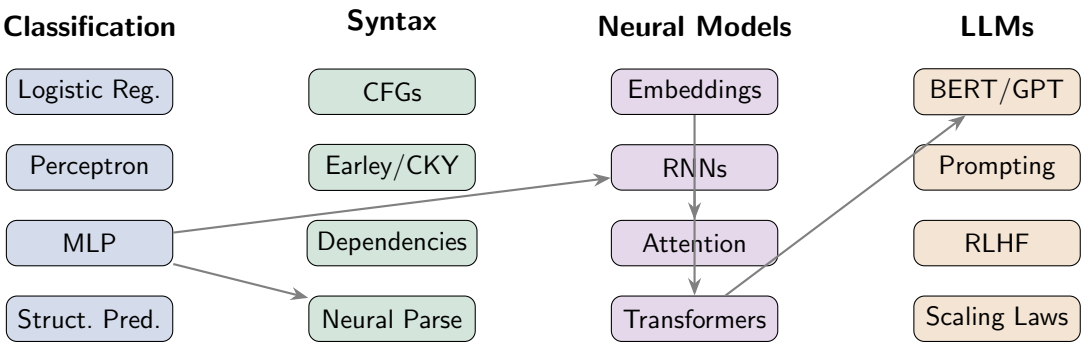
Key skills: Word embeddings, CFG parsing, dependency structures, neural parsers.

Schedule: Neural Models & LLMs (Weeks 9–15)

Date	Topic	Readings
3/31	Recurrent Neural Networks	JM19
4/10	RNNs and Transformers	JM8
4/14	Transformers; Positional Encoding	JM9
4/17	Large Language Models	JM10
4/21	Prompting; Prompt Engineering	JM11; JM12
4/24	Human Preference; RLHF; Scaling Laws	—
4/28	**Quiz 2**	—

Key skills: RNN architectures, attention mechanisms, transformers, LLM training, prompting, RLHF.

Key Topics: A Visual Overview



Coming Up: Two Cultures of Linguistics

Our first topic: The relationship between linguistic theory and data-driven NLP.

Rationalist Tradition:

- Formal grammars
- Symbolic representations
- Linguistic universals
- Chomsky, Montague

Empiricist Tradition:

- Statistical patterns
- Learning from data
- Distributional semantics
- Neural networks, LLMs

Key question: How do these traditions inform modern NLP?

This course bridges both: formal linguistic analysis meets machine learning.

What You'll Build

By the end of this course, you will have implemented:

1. **Text classifiers** using logistic regression and neural networks
2. **Sequence labelers** (POS taggers, NER) using structured perceptrons
3. **Parsers** for syntactic analysis (constituency and dependency)
4. **Word embeddings** and distributional semantic models
5. **Neural language models** including RNNs and transformers
6. Experience with **prompting and fine-tuning** LLMs

All implemented in Python/PyTorch—real, working code you can build on.

Advice for Success

Do:

- Start assignments early
- Read before class
- Ask questions (office hours, forum)
- Form study groups
- Implement things yourself first
- Debug systematically

Don't:

- Wait until the last minute
- Skip the math—it matters
- Copy code without understanding
- Ignore the readings
- Suffer in silence

The goal: Deep understanding, not just surface familiarity.

Let's begin!

Two Cultures of Linguistics

Linguistics in the Age of Big Data and LLMs

(next slide deck)